MEDINOVA TEACHING HOSPITAL

Class 12 — Infectious Diseases

Sepsis · Stewardship · Tropical Medicine · HIV

> *"Know your enemy before you prescribe your weapon."*

FELLOWSHIP SERIES | MODULE 12 of 20

SECTION 01

Antimicrobial Stewardship (AMS)

AMS aims to optimize clinical outcomes while minimizing unintended consequences of antimicrobial use (resistance, toxicity, and *C. diff*).

The ESKAPE Pathogens: (High priority for resistance)
***E**nterococcus faecium*, ***S**taphylococcus aureus*, ***K**lebsiella pneumoniae*, ***A**cinetobacter baumannii*, ***P**seudomonas aeruginosa*, ***E**nterobacter* spp.

IV to Oral Switch Criteria (COMS)

- **C - Clinical:** Improving clinical signs and symptoms.
- **O - Oral:** Patient able to tolerate oral intake (no vomiting/malabsorption).
- **M - Markers:** Improving inflammatory markers (Temp < 38°C for 24h, WBC falling).
- **S - Specific:** No "deep-seated" infections requiring prolonged IV (e.g., Endocarditis, Osteomyelitis).

> **Prof. Obi's Pearl #1** "Starting an antibiotic is a clinical decision; stopping one is a professional responsibility. If you don't de-escalate based on cultures after 48 hours, you are practicing lazy medicine."

SECTION 02

Common Antibiotic Classes Comparison

Class	Mechanism	Spectrum	Monitoring / Side Effects
Beta-Lactams (Pen/Ceph)	Cell wall synthesis inhibition	G+ve, G-ve (varies by gen)	Hypersensitivity (Anaphylaxis). *C. diff* risk.
Aminoglycosides (Gent)	30S Ribosome inhibition	Aerobic G-ve bacilli	**Levels required.** Ototoxicity & Nephrotoxicity.
Glycopeptides (Vancomycin)	Cell wall (D-Ala-D-Ala)	G+ve (MRSA)	**Trough levels.** Red Man Syndrome.
Quinolones (Cipro)	DNA Gyrase inhibition	Broad (inc. Pseudomonas)	Tendon rupture. Prolonged QTc. CNS effects.
Macrolides (Azithro)	50S Ribosome inhibition	Atypicals (Legionella/Chlamy)	GI upset. Liver enzyme inhibition.

Prof. Obi's Pearl #2 "Vancomycin trough levels are not a suggestion; they are a safety requirement. Under-dosing breeds resistance; over-dosing kills the kidneys. Pick one."

SECTION 03

Tropical Infections: The Nigerian Context

WHO Criteria for Severe Malaria

Presence of *P. falciparum* parasitaemia plus one or more of:

- Impaired consciousness (Cerebral Malaria).
- Severe Anaemia (Hb < 50 g/L).
- Acute Kidney Injury (Creatinine > 265 µmol/L).
- Pulmonary Oedema / ARDS.
- Hypoglycaemia (< 2.2 mmol/L).
- Jaundice + high parasitaemia.

Management: IV Artesunate (2.4 mg/kg at 0, 12, 24h) is the gold standard.

Lassa Fever (Viral Haemorrhagic Fever)

- **Presentation:** Fever, sore throat, retrosternal pain, facial swelling, mucosal bleeding.
- **Action:** Immediate Isolation + Notify Public Health + IV Ribavirin (if early).

SECTION 04

HIV & Opportunistic Infections

CD4 Count	Pathogen / Condition	Prophylaxis
Any	TB / Kaposi Sarcoma	Screen for TB / Isoniazid
< 200	Pneumocystis (PJP)	Co-trimoxazole (Septrin)
< 100	Toxoplasmosis / Cryptococcus	Co-trimoxazole / Fluconazole
< 50	Mycobacterium Avium (MAC)	Azithromycin

SECTION 05

Healthcare-Associated Infections (HCAI)

1. **MRSA:** Rx: Vancomycin or Linezolid. Decolonization with Mupirocin.
2. **C. difficile:** Profuse diarrhea post-antibiotics. Rx: Oral Vancomycin (125mg QDS) or Fidaxomicin.
3. **VRE:** Vancomycin-Resistant Enterococcus. Requires strict contact isolation.

> **Prof. Obi's Pearl #3** "In HIV, the Immune Reconstitution Inflammatory Syndrome (IRIS) can make a recovering patient look like they are dying. Paradoxical worsening is a sign the immune system is finally waking up."

SECTION 06

Infection Control & Precautions

Precaution	Examples	Personal Protective Equipment (PPE)
Standard	Every patient	Gloves/Apron if body fluid risk. Hand hygiene.
Contact	MRSA, *C. diff*, Norovirus	Gloves + Gown. Dedicated equipment.
Droplet	Meningitis, Influenza, Pertussis	Surgical Mask. Eye protection.
Airborne	TB, Measles, Chickenpox	FFP3/N95 Mask. Negative pressure room.

SECTION 07

Fungal Infections Quick Reference

- **Candida:** (Albicans/Non-albicans). Rx: Echinocandins (Anidulafungin) or Fluconazole.
- **Cryptococcus:** (Meningitis in HIV). Dx: India Ink or CRAG. Rx: Amphotericin B + Flucytosine.
- **Aspergillus:** (Galactomannan +ve). Rx: Voriconazole.

> **Prof. Obi's Pearl #4** "Hand hygiene is the single most effective 'drug' we have against hospital-acquired infections. It costs nothing and saves more lives than our most expensive carbapenems."

OFFICIAL SIGN-OFF

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INSTITUTION: MediNova Teaching Hospital
SERIES: Fellowship Module 12 of 20
DATE: March 2026